AIRSTREET · STEM CHECK

AirStreet STEM check

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-24	Amish Chadha	First STEM check (kit 1.9.1 upgrade) of the current design (option A) and options B and C; four rows failed with a fix and were fixed under Amish's standing instruction of 2026-10-23 (AST-DDR-005); independent review recorded

AirStreet STEM check

Verdict: the working principle of the current design, option A, sits within known physics, chemistry and biology, and no row fails outright. Four details failed on first check and were fixed in this session under Amish's standing instruction of 2026-10-23 (AST-DDR-005): the NO2 noise term was below the sensor data sheet (S10), the ozone filter of the NO2 cell can be used up before the cell's two-year replacement (S14), readings outside the sensors' rated ranges were not marked (S15), and the model drew a thicker band than the calculation used (S24). One row stays **Fails, fix proposed** and goes to Amish because its fixes change the user-facing scope: the NO2 cell is rated only to 40 degC and 85 % RH, while the node is built for 50 degC and humid nights (S33; open decision 3). Eleven rows of option A are plausible, to confirm at TRL 4 or in FieldNode, among them the field NO2 accuracy (S11, now about 4.3 ppb hourly after the independent review), the PM humidity correction (S16), the shield's radiation error (S17), the band at its worm slots (S23), the insert pull-out (S26) and the sampling of the hourly means (S34). So `stem_check` is **open**. Of 42 claims across the three options, 29 are within known science, 12 are plausible and 1 fails with a fix proposed to Amish; options B and C have no failing row. Method: STANDARDS section 20. Calculations: `docs/04-calcs/stem_check.py`, which runs `docs/04-calcs/sizing.py` (AST-CAL-001 v0.7) for every option A number.

Claims checked

Table 1. Claims, governing science and verdicts (generated from `docs/04-calcs/stem_results.csv`). Rows S1 to S32 are option A, B1 to B4 option B and C1 to C3 option C.

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
S1	The sensor port load (design case, maximum SPS30 current and 1.2 margin) fits FieldNode's 100 mW design allowance	Physics	Electromagnetism and circuits	P = V I x duty; SPS30 datasheet 65 mA maximum at 5 V, 60 s in 300 s; front end 1.5 mA held on (CAL [A3])	87 mW <= 100 mW	Within known science	AST-CAL-001
S2	In FieldNode's worst month the cell is charged with more energy than the node draws in a day	Chemistry	Electrochemistry and batteries	Energy balance; FND-CAL-001 worst-month storage 7.75 Wh/day (CAL [A7])	in 7.75 Wh/day, out 2.33 Wh/day	Within known science	AST-CAL-001
S3	A full 6 Ah LiFePO4 cell (80 % usable) runs the node 6.6 days without sun	Chemistry	Electrochemistry and batteries	Usable energy over average draw (97 mW) (CAL [A6])	claimed 6.6 days, estimate 6.6 days (ratio 1.00)	Within known science	AST-CAL-001
S4	The LiFePO4 cell is not charged below 0 degC; a cold spell is ridden through on the cell (4.6 days at -20 degC)	Chemistry	Electrochemistry and batteries	LiFePO4 charge temperature limit (cell maker's data, in FieldNode); capacity ratio 0.70 at -20 degC (FND-CAL-001)	Autonomy 4.6 days at -20 degC (CAL [A6]); the first partner's climate data must show no sub-zero spell longer than that	Plausible, to confirm at TRL 3	Open
S5	A 20-byte uplink takes about 247 ms at SF9 on a 125 kHz channel	Engineering and technology	Power and link budgets	Semtech AN1200.13 time-on-air formula, 33 bytes with LoRaWAN overhead, 48 payload symbols (CAL [B2])	claimed 247 ms, estimate 247 ms (ratio 1.00)	Within known science	AST-CAL-001
S6	The worst-case uplink (SF12) stays under the 1 % duty cycle of an EU868 sub-band	Engineering and technology	Standards practice	ETSI EN 300 220 and LoRaWAN EU868 regional parameters; 1.81 s in each 300 s (CAL [B3])	0.6 % <= 1 %	Within known science	AST-CAL-001
S7	In US915 the 20-byte record fits the 400 ms		Standards practice	FCC 47 CFR 15.247 dwell time; LoRaWAN US915 regional	247 ms <= 400 ms		AST-CAL-001

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	dwell limit at SF9 (DR1 carries up to 53 bytes; DR0 at SF10 carries 11)	Engineering and technology		parameters (CAL [B5])		Within known science	
S8	At 15 min on The Things Network the node keeps within 30 s of airtime a day up to SF9	Engineering and technology	Standards practice	The Things Network fair use policy, 30 s per device per day (CAL [B8])	23.7 s/day <= 30 s/day	Within known science	AST-CAL-001
S9	The 16-bit ADC's step is not what limits the NO2 reading (the cell's noise is)	Chemistry	Analytical	NO2-B43F sensitivity -200 to -650 nA/ppm (Alphasense datasheet); 0.8 mV/nA front end; ADS1115 62.5 uV step	0.16 to 0.52 mV/ppb, so one ADC step is 0.12 to 0.39 ppb, 20 to 60 times below the 7.5 ppb noise per sample (CAL [C1])	Within known science	AST-CAL-001
S10	Noise term of the hourly NO2 budget: per-sample noise taken from the sensor datasheet	Mathematics	Statistics and uncertainty	NO2-B43F datasheet noise 15 ppb at 2 sigma; averaging 120 samples an hour divides it by the square root of 120	Was 5 ppb (1 sigma) assumed over 120 samples (Fails, fix proposed); now 7.5 ppb over the 12 independent records of an hour (the 10 samples of a record fall within the cell's response time), 2.17 ppb hourly, total 5.45 ppb, MAE 4.3 ppb (CAL [C2], [C3])	Within known science	AST-DDR-005 F1
S11	After field calibration the hourly NO2 mean absolute	Mathematics	Statistics and uncertainty	Field-calibrated residual 3.5 ppb MAE (Zimmerman et al., 2018, a model that	Budget 5.45 ppb (1 sigma), MAE 4.3 ppb; only	Plausible, to confirm at TRL 3	Open

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	error is about 4.3 ppb, within 5 ppb			also drew on other gas sensors on the node, so optimistic for WE, AE, T and RH alone); drift, street transfer bias and a 5 % slope error assumed (CAL C)	collocation at TRL 4 can show it (PR-002 at risk)		
S12	An annual NO2 mean cannot be resolved to 2 ug/m3, so AirStreet makes no claim at the WHO guideline	Mathematics	Statistics and uncertainty	Systematic terms do not average out over a year (CAL [C4])	4.65 ug/m3 >= 2 ug/m3	Within known science	AST-CAL-001
S13	1 ppb of NO2 is 1.88 ug/m3 at 25 degC, and 1.91 ug/m3 at the 20 degC the EU directive uses for limit values	Chemistry	Environmental and atmospheric chemistry	Ideal gas molar volume 24.465 L/mol at 25 degC and 24.055 L/mol at 20 degC; NO2 46.006 g/mol; Directive 2008/50/EC refers gas volumes to 293 K (CAL [C1], [C5])	claimed 1.913 ug/m3 per ppb at 20 degC, estimate 1.91 ug/m3 per ppb at 20 degC (ratio 1.00)	Within known science	AST-CAL-001
S14	The NO2 cell's ozone filter lasts until the cell is replaced	Chemistry	Analytical	NO2-B43F datasheet: ozone filter capacity more than 500 ppm h at 0.5 ppm O3; operating life more than 24 months (to 50 % signal)	At 40 to 60 ppb mean ozone the filter is used up in 17 to 11 months, before the 24-month replacement assumed before (Fails, fix proposed); now the cell is replaced every 12 months at the pod exchange, or at 18 or 24 months where the site's ozone record keeps the dose under 500 ppm h (CAL [C7])	Within known science	AST-DDR-005 F3

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
S15	Every reading taken outside the sensors' rated temperature and humidity ranges is marked as such in the data	Engineering and technology	Sensing resolution	NO2-B43F rated -30 to 40 degC and 15 to 85 % RH continuous; SPS30 recommended 10 to 40 degC and 20 to 80 % RH (datasheets)	Rated-range flags in every record: shield air above 35 degC (a 5 K pod allowance, to measure at TRL 4), RH above 80 % or 85 % and below 20 % or 15 %, below 10 degC for PM; flagged hours are published with the flag (CAL [H5])	Within known science	AST-DDR-005 F2
S16	After humidity correction the hourly PM2.5 error stays within 5 ug/m3 or 30 % up to 80 % RH	Physics	Optics and radiation	Laser scattering counts particles; kappa-Koehler water uptake 1 + kappa aw / (1 - aw), kappa 0.3 +/- 0.1 assumed (CAL E)	Within the bound to 80 % RH, outside it at 85 % RH and above at 25 and 50 ug/m3 (CAL [E2]); the comparison is at 1 sigma (about 68 % of hours), the base error and kappa are assumed and the RH is the shield's, not the pod's, so collocation at TRL 4 confirms it	Plausible, to confirm at TRL 3	Open
S17	The passive eight-plate shield keeps the radiation error of the air temperature under 1 K at 1 m/s wind in full sun	Physics	Heat transfer	Plate energy balance and probe convection (Zukauskas), inner air at 10 to 30 % of the wind (assumed) (CAL D)	0.87 K <= 1 K	Plausible, to confirm at TRL 3	Open

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
S18	A 0.87 K warm error reads about 4.1 % RH low at 85 % RH and 30 degC	Physics	Thermodynamics	Magnus saturation vapour pressure (Sonntag constants) (CAL [D3])	claimed 4.1 % RH, estimate 4.13 % RH (ratio 0.99)	Within known science	AST-CAL-001
S19	A 35 m/s gust puts about 125 N on the node	Physics	Fluid mechanics	F = 0.5 rho v^2 Cd A, Cd 1.2 on the 0.139 m2 model silhouette; 35 m/s is a screening speed, 206 N at 45 m/s for windier codes (CAL [G1])	claimed 125 N, estimate 125 N (ratio 1.00)	Within known science	AST-CAL-001
S20	Two bands hold the node against slipping down the pole	Physics	Mechanics and statics	Belt friction F = mu T theta per band, 1,000 N preload and mu 0.3 assumed; load is weight and wind on the tilted panel (CAL [G2])	75 N <= 1885 N	Within known science	AST-CAL-001
S21	The 40 x 4 mm rail at the lower clamp stays below the yield of 6063 aluminium with a factor of 1.5, in bending and twist together	Physics	Strength of materials	Bending M / Z and flat-bar torsion T / (0.31 b t^2), von Mises; 6063-T5 minimum yield about 110 MPa (CAL [G6], [G8])	33.2 MPa <= 73 MPa	Within known science	AST-CAL-001
S22	The cross arm carries the pod and the shield far below yield (with a factor of 1.5)	Physics	Strength of materials	Combined bending of a 30 x 30 x 3 angle (CAL [G3])	4.4 MPa <= 100 MPa	Within known science	AST-CAL-001
S23	The 12.7 x 0.6 mm stainless band carries its 1,000 N preload below yield (with a factor of 1.5)	Physics	Strength of materials	Tension stress T / (w t); 304 stainless, minimum yield 205 MPa annealed (ASTM A240); worm-drive bands are cold rolled and stronger, but slotted	Gross section 131 MPa against 137 MPa; at the worm slots, with about 35 % of the section cut away, about 200 MPa; to confirm with the band	Plausible, to confirm at TRL 3	Open

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
					maker's rated tension (AST-DEC-001, to confirm item 1)		
S24	The band thickness used for the mass is the one in the model	Mathematics	Units and dimensional analysis	One source of dimensions: the calculation reads PARAMS (STANDARDS section 24)	The model drew a 0.8 mm band while the mass used 0.6 mm (Fails, fix proposed); now 0.6 mm in the model and read from it (CAL [F4])	Within known science	AST-DDR-005 F4
S25	Printed ASA saddles keep the band preload over years in sun	Chemistry	Materials and polymer chemistry	ASA is UV stable outdoors; printed thermoplastics creep under sustained load, faster when warm (heat deflection of ASA about 95 degC at 0.45 MPa)	Contact stress on the V faces is low (a few MPa), but creep at up to about 60 degC surface temperature can relax the preload; a re-torque at the first pod exchange and a check at TRL 4	Plausible, to confirm at TRL 3	Open
S26	The pod's two hanging screws hold in their heat-set inserts with a wide margin	Physics	Strength of materials	Worst pull-out per screw from weight and wind prying about the 40 mm bearing edge, shared by the two screws in line (CAL [G7]), against an M5 insert in printed ASA	About 11 N per screw against an assumed 200 N for the insert, which is not yet from maker data and falls when the ASA is warm; pull test at 23 and 60 degC at TRL 4 (AST-DEC-001, to confirm item 8)	Plausible, to confirm at TRL 3	Open
S27	Traffic NO2 varies several fold within city blocks, so	Biology	Epidemiology	Apte et al. (2017), Oakland: NO2 varied 5 to 8 times within single blocks; WHO	Cited as the problem, not as a health claim; AirStreet	Within known science	Open

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	street-level nodes see what a city station misses			ambient air quality fact sheet	gives indicative levels, not health advice		
S28	Inlets 3.0 m above the sidewalk sample the street air people breathe closely enough to rank streets	Chemistry	Environmental and atmospheric chemistry	EU Directive 2008/50/EC Annex III: inlets 1.5 to 4 m above ground; Defra diffusion tube guidance: 2 to 4 m, higher tubes may read low	Inside the regulated range; a small vertical gradient is part of the transfer bias term (CAL C)	Within known science	Open
S29	A 60 s particle run leaves at least 30 s of valid readings in clean air	Engineering and technology	Sensing resolution	SPS30 datasheet: up to 30 s to the first reading at low counts (CAL [A9])	30 s >= 30 s	Within known science	AST-CAL-001
S30	The node with FieldNode's sun shield stays within 4.0 kg on the pole	Engineering and technology	Manufacturing tolerances	Part volumes from the model and densities (CAL [F5])	3.77 kg <= 4 kg	Within known science	AST-CAL-001
S31	Stainless insect mesh under the pod keeps insects and spiders out of the sensors	Biology	Ecology and agronomy	Insect screens stop insects larger than the aperture; spiders and wasps nest in sheltered openings	Mesh aperture to be chosen for the smallest local insects; nesting is checked at each pod exchange (TRL 4)	Plausible, to confirm at TRL 3	Open
S32	FieldNode's core survives 50 degC ambient with its sun shield fitted above 45 degC	Physics	Heat transfer	FieldNode is rated to 45 degC ambient and runs 58.7 to 73.3 degC inside at 45 degC (FND-CAL-001); the sun shield is FieldNode's fix	To confirm in FieldNode's calculation note; AirStreet relies on it at hot sites	Plausible, to confirm at TRL 3	Open
S33	The NO2 cell and the particle sensor measure validly across the whole climate the node is	Engineering and technology	Sensing resolution	NO2-B43F rated -30 to 40 degC and 15 to 85 % RH continuous; SPS30 recommended 10 to 40 degC and 20 to 80 % RH (datasheets)	Above 40 degC and 85 % RH the cell runs outside its rating; the flags (S15) mark those hours	Fails, fix proposed	Open

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
	built for (-10 to 50 degC, 10 to 100 % RH)				but do not make them valid. Fixes that change the user-facing scope (a wider-range cell, or a narrower PR-012) go to Amish (AST-DEC-001, open decision 3)		
S34	Twelve 10 s NO2 readings and twelve 60 s PM runs an hour represent the hourly mean	Mathematics	Statistics and uncertainty	Sampling a fluctuating signal: the error of the hourly mean falls with the number of independent samples; the cell's response (T90 under 80 s) smooths the NO2 signal	Street NO2 changes over seconds near traffic; the sampling error is not in the budget and is measured in the TRL 4 collocation against the reference's continuous record	Plausible, to confirm at TRL 3	Open
S35	The printed and bolted parts survive years of wind vibration and occasional ice	Physics	Acoustics and vibration	Fatigue under vortex shedding and gusts; ice load on the shield and panel	Stresses are low (S21, S22), but no fatigue or ice case is calculated; nyloc nuts and inserts resist loosening; checked at TRL 4 and by inspection at each pod exchange	Plausible, to confirm at TRL 3	Open
B1	Option B: a 30 W panel charges more than the heated, pumped node uses in a winter day	Engineering and technology	Power and link budgets	Energy balance with estimated loads (pump 1 W at 20 %, heater 2 W for 30 % of hours, fan 0.25 W, sensors 0.1 W)	in 45 Wh/day, out 27.6 Wh/day	Within known science	Open

#	Claim	Field	Branch	Governing law or source	First-order check	Verdict	Closed by
B2	Option B: warming the 1 L/min sample by 5 K takes far less than the 2 W heater gives	Physics	Thermodynamics	P = m_dot cp dT; most of the heater power is lost through the sleeve	0.1 W <= 2 W	Within known science	Open
B3	Option B: warming the sample 5 K brings 90 % RH below the SPS30's 80 % recommended limit	Physics	Thermodynamics	Magnus saturation vapour pressure at 20 and 25 degC	66 % RH <= 80 % RH	Within known science	Open
B4	Option B: the heated inlet does not change the particles it measures	Chemistry	Environmental and atmospheric chemistry	Semi-volatile ammonium nitrate evaporates when an inlet is heated (known loss in heated-inlet PM monitors)	A few kelvin of warming loses part of the nitrate fraction; to quantify by collocation if B were chosen	Plausible, to confirm at TRL 3	Open
C1	Option C: a Palmes tube collects about 1.5 ug of NO2 in 28 days at 30 ug/m3	Chemistry	Analytical	Fick's first law, uptake D A / L with D(NO2) 0.154 cm2/s, 11 mm bore, 71 mm long	claimed 1.5 ug, estimate 1.5 ug (ratio 1.00)	Within known science	Open
C2	Option C: triplicate tubes give monthly NO2 means within about 25 %	Mathematics	Statistics and uncertainty	Defra NO2 diffusion tube practical guidance: indicative technique, uncertainty about 25 %; bias adjusted by collocation with a chemiluminescence analyser	Monthly means only; no hourly or daily data	Within known science	Open
C3	Option C: four alkaline D cells run the particle logger for a month between tube changes	Chemistry	Electrochemistry and batteries	Energy balance; alkaline D capacity about 12 Ah at low drain (estimate)	in 50.4 Wh, out 14.6 Wh	Within known science	Open

Verdicts: **Within known science**, **Plausible, to confirm at TRL 3**, **Fails, fix proposed**, **Fails**. A row closed by a fix names the decision record in "Closed by" and states in its check what failed before.

Fixes made under the standing instruction

Table 2. Rows that failed on first check and the fix made on 2026-10-24 (AST-DDR-005, AST-DEC-001), with the corrections the independent review added the same day. None changes the working principle, the pitch or the user-facing scope; Amish can reverse each.

Row	What failed	Fix
S10	The calculation note assumed 5 ppb (1 sigma) of noise per NO2 sample over 120 samples an hour; the NO2-B43F data sheet gives 15 ppb at 2 sigma (7.5 ppb), and the review showed that only the 12 records of an hour are independent	F1: AST-CAL-001 uses 7.5 ppb over 12 records (2.17 ppb hourly) and adds a 5 % slope term; the hourly total rises from 4.92 to 5.45 ppb and the mean absolute error from 3.9 to 4.3 ppb, still under 5 ppb
S14	The cell's ozone filter holds 500 ppm h: 17 months at 40 ppb mean ozone and 11 to 14 months at the 50 to 60 ppb of hot, sunny sites, before the two-year replacement assumed	F3: the cell is replaced every 12 months at the pod exchange, or at 18 or 24 months where the collocated reference station's ozone record keeps the dose since fitting under 500 ppm h
S15	Readings taken outside the sensors' rated ranges (NO2-B43F to 40 degC and 15 to 85 % RH; SPS30 recommended 10 to 40 degC and 20 to 80 % RH) were not marked	F2: every record carries rated-range flags (shield air above 35 degC, with a 5 K pod allowance to measure at TRL 4; RH above 80 % or 85 % and below 20 % or 15 %; below 10 degC for PM); flagged hours are published with the flag. Validity above the rated range is S33, for Amish
S24	The model drew the band 0.8 mm thick while the mass used 0.6 mm	F4: the model's band is 12.7 x 0.6 mm, the common worm-drive band, and the calculation reads it from the model

Fields not relevant

Every field is touched. Biology enters only through the problem statement (S27) and insects in the pod (S31): AirStreet gives indicative pollutant levels and makes no health claim (AST-PRC-001, Safety).

Concept options

- Option A, electrochemical and optical node on FieldNode (current design, AST-OPT-001): the rows S1 to S35; four fixes made, eleven rows plausible, one (S33) with its fix proposed to Amish.
- Option B, aspirated, heated-inlet sampling node: within known science on energy (B1), heating (B2) and drying (B3); heating the sample loses part of the semi-volatile nitrate in the particles (B4, plausible), a known effect of heated inlets.
- Option C, passive diffusion tubes with a battery PM logger: within known science; Palmes tubes collect about 1.5 ug of NO2 a month (C1) and give monthly means within about 25 % (C2), and four D cells run the PM logger for a month (C3). Its limit is that it gives monthly NO2 means only, which is a requirement shortfall (AST-OPT-001), not a science failure.

Independent review

On 2026-10-24 a second reviewer (a separate Claude session run headless, `claude -p`, in a folder holding only this check, AST-CAL-001, AST-PRS-001 and `sizing.py`, with no access to the author's reasoning) recomputed the headline results and tried to refute every "Within known science" verdict. It reproduced the power budget, airtime, autonomy, wind load, clamp slip, band, rail and mass figures, and refuted or weakened the rows below. Each finding was checked in the sizing script before anything changed.

Table 3. Independent review findings and what changed.

Row or result	Finding	Verified	Change
S10, C2	The 10 samples of a record fall within the cell's response time, so the noise averages over about 12 independent values an hour, not 120	Yes	Noise term 7.5 ppb over 12 records (2.17 ppb); hourly total 5.45 ppb with the slope term; MAE 4.3 ppb, under 5 ppb (PR-002 stays at risk)
S11, C2 to C5	No span (slope) term; the 3.5 ppb residual comes from a model that also used other gas sensors, so it is optimistic for WE, AE, T and RH alone	Yes (slope); the source's inputs are noted, not re-read	5 % slope term at a 20 ppb street mean (1.0 ppb, assumed); annual 2.47 ppb (4.65 ug/m3); S11 wording names the caveat; stays plausible
S12, S13, C5	The EU directive refers gas limits to 20 degC, where 1 ppb is 1.913 ug/m3, not 1.880	Yes	C5 at 20 degC: decisive below 33.9 or above 46.1 ug/m3 against 40, and 13.9 to 26.1 against the 2030 limit of 20
C5	The 2 ppb street transfer bias may be optimistic (a 5 to 10 % fall in NO2 from 1.5 to 3 m)	Partly (assumption, not measured)	Sensitivity line C8: with 3 ppb the 40 ug/m3 band is 31.8 to 48.2 ug/m3; measured at TRL 4
S14	40 ppb is not a high ozone case; hot sunny sites run 50 to 60 ppb (11 to 14 months); not every site has an ozone record	Yes	F3 changed to a 12-month replacement, extended to 18 or 24 months only where the ozone record allows
S15	The verdict was closed by rewriting the check, not the claim; the 3 K pod allowance is unsupported; no low-RH flags	Yes	S15 restated as the flagging claim (closed by F2); new row S33 keeps the original claim as Fails, fix proposed, for Amish; pod allowance 5 K, to measure at TRL 4; low-RH flags added
S16	The bound is compared at 1 sigma (about 68 % of hours); base error and kappa range narrow; shield RH, not pod RH	Yes	Stated in S16; stays plausible; PR-001 stays at risk
S17, D	The 0.87 K passes 1 K once the inner air falls below about 8 % of the wind	Yes: 8.3 %	Line D4 added; S17 stays plausible
S8, B	The TTN payload scheme was not stated; a 7-day backlog cannot be flushed on TTN	Yes	Each 15 min TTN uplink carries the mean of three 5 min records; line B9: a backlog takes about 14 h at SF9 through a private gateway and cannot be flushed on TTN
G4	The bolt couple was halved: the worse cross arm bolt takes about 127 N, not 67 N	Yes	G4 corrected; margin against the 7,000 N proof load still large
S21, G6	The yield used was typical (150 MPa), not minimum (about 110	Yes	

Row or result	Finding	Verified	Change
	MPa); the arm's twist was not combined		G8: 12.2 MPa torsion, 33.2 MPa von Mises against 73 MPa
S23	The band's net section at the worm slots (about 35 % less) reaches about 200 MPa	Yes, as an estimate	S23 plausible until the band maker's rated tension is checked (AST-DEC-001, to confirm item 1)
S26, G7	The 200 N insert strength is unverified and falls when warm; the reviewer read the prying term as halved	The 200 N point yes; the prying term no: the two screws lie in one line along the arm, so they share the prying tension about the 40 mm bearing edge, as the script has it	S26 plausible until a pull test at 23 and 60 degC
S2, S4	Worst-month charging assumes the cell charges; cold and end of life were not combined	Yes	A6 adds 3.7 days cold at end of life; S4 names the charging rule
S19	35 m/s is a screening speed	Yes	G1 adds 206 N at 45 m/s
S9	The row proved nothing about resolution	Yes	Reworded: the ADC step is not the limit
Requirement counts	The PRS counted five met by design; there are six	Yes	AST-PRS-001 corrected
Missing checks	Sampling of the hourly means; fatigue and ice; pod airflow and condensation; span drift	Yes	S34 and S35 added (plausible); pod airflow and condensation stay under PR-012 (at risk) and S31

Upheld without change: S1, S3, S5 to S7, S18, S20, S22, S27 to S30.

Sources

- Alphasense, NO2-B43F nitrogen dioxide sensor data sheet (sensitivity -200 to -650 nA/ppm, noise 15 ppb at 2 sigma, ozone filter more than 500 ppm h, operating -30 to 40 degC and 15 to 85 % RH continuous, life more than 24 months), https://pdf.directindustry.com/pdf/alphasense/no2-b43f/16860-1025996.html, and product page https://www.alphasense.com/products/view-by-target-gas/no2-b43f, both accessed 2026-10-24.
- Sensirion, SPS30 data sheet (07/2023): 55 mA typical and 65 mA maximum in measurement mode, up to 30 s to first reading, recommended 10 to 40 degC and 20 to 80 % RH, https://sensirion.com/products/catalog/SPS30, accessed 2026-09-25.
- Semtech AN1200.13, LoRa modem designer's guide (time on air); The Things Network, "Regional limitations of RF use" (EU868 1 % duty cycle, US915 400 ms dwell time), https://www.thethingsnetwork.org/docs/lorawan/regional-limitations-of-rf-use/, and "The Things Network Fair Use Policy" (30 s of uplink airtime a day), https://www.thethingsnetwork.org/forum/t/the-things-network-fair-use-policy/47689, both accessed 2026-10-24; the US915 DR0 payload of 11 bytes as cited in SLW-REG-001, to confirm against the LoRaWAN regional parameters text.
- C. Zimmerman et al., "A machine learning calibration model using random forests to improve sensor performance for lower-cost air quality monitoring," *Atmospheric Measurement Techniques* 11 (2018): 291 to 313, https://amt.copernicus.org/articles/11/291/2018/.

- K. K. Barkjohn et al., "Development and application of a United States-wide correction for PM2.5 data collected with the PurpleAir sensor," *Atmospheric Measurement Techniques* 14 (2021): 4617 to 4637, https://amt.copernicus.org/articles/14/4617/2021/.
- J. S. Apte et al., "High-resolution air pollution mapping with Google Street View cars," *Environmental Science and Technology* 51 (2017): 6999 to 7008, https://pubs.acs.org/doi/10.1021/acs.est.7b00891.
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- FieldNode calculation note FND-CAL-001 (v0.1 power and energy, v0.3 mass) and FieldNode requirements FND-REQ-001, in the fieldnode repository.
- AST-CAL-001 v0.7 for every option A number.